Note

If two lines are skew lines, then they are non coplanar.

If the lines are not parallel and intersect, the distance between them is zero. If they are parallel and non-intersecting, the distance is determined by the length of the line segment perpendicular to both the parallel lines. In the same way, the shortest distance between two skew lines is defined as the length of the line segment perpendicular to both the skew lines. Two lines will either be parallel or skew.

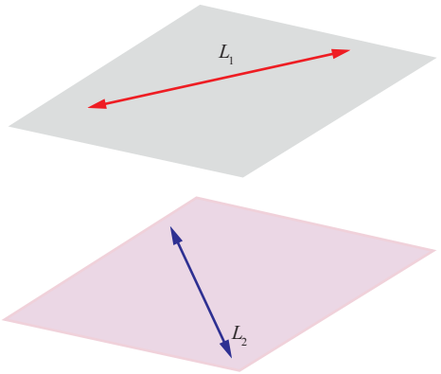


Fig. 6.20

Theorem 6.13

The shortest distance between the two parallel lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{b}$ is given by $d = \dfrac{|(\vec{c} - \vec{a}) \times \vec{b}|}{|\vec{b}|}$, where $|\vec{b}| \neq 0$.

Proof

The given two parallel lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{b}$ are denoted by L_1 and L_2 respectively. Let A and B be the points on L_1 and L_2 whose position vectors are $\vec{a}$ and $\vec{c}$ respectively. The two given lines are parallel to $\vec{b}$.

Let AD be a perpendicular to the two given lines. If θ is the acute angle between $\overrightarrow{AB}$ and $\vec{b}$, then

$$\sin\theta = \frac{|\overrightarrow{AB} \times \vec{b}|}{|\overrightarrow{AB}|\,|\vec{b}|} = \frac{|(\vec{c} - \vec{a}) \times \vec{b}|}{|\vec{c} - \vec{a}|\,|\vec{b}|} \qquad \ldots (1)$$

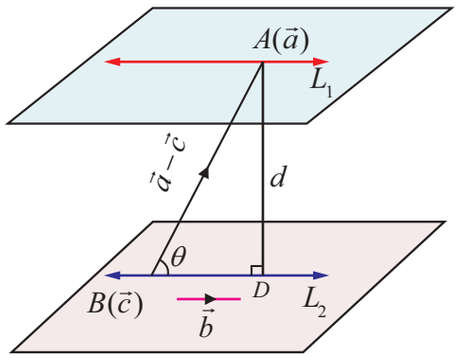


Fig. 6.21

But, from the right angle triangle ABD,

$$\sin\theta = \frac{d}{AB} = \frac{d}{|\overrightarrow{AB}|} = \frac{d}{|\vec{c} - \vec{a}|} \qquad \ldots (2)$$

From (1) and (2), we have $d = \dfrac{|(\vec{c} - \vec{a}) \times \vec{b}|}{|\vec{b}|}$, where $|\vec{b}| \neq 0$.

Theorem 6.14

The shortest distance between the two skew lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ is given by

$$\delta = \frac{|(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|}, \text{ where } |\vec{b} \times \vec{d}| \neq 0$$

Proof

The two skew lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ are denoted by L_1 and L_2 respectively.

Let A and C be the points on L_1 and L_2 with position vectors $\vec{a}$ and $\vec{c}$ respectively.

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